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Locating resolution-dependent behavioral correspondence in qualitative DNA-damage response models

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Qualitative DNA-damage response models can share functional organization while encoding different regulatory and terminal mechanisms. Their correspondence therefore depends on which biological events a comparison observes. We ask at what declared resolution curated animal/human and *Arabidopsis* models cease to support matching damage-response choices. We construct finite executable transition systems from the models and compare their recursive branching under three literature-motivated observational interfaces. The two unchanged Petri-net structures have PN-GDDA similarity 0.9976. At coarse functional and pathway-level resolution, the models match each other's observable steps and choices after internal events are hidden: they are weakly bisimilar. Distinguishing mammalian apoptosis from the encoded plant SMR and differentiation/endoreduplication sequence eliminates both simulation directions, while structural similarity remains constant. This comparison locates a compatibility boundary within the declared interfaces, rather than discovering the already encoded mechanistic differences. Formal decisions are supported by proofs and independent computational checks. The result shows how readout selection changes conclusions about these executable models; it does not establish organism-level equivalence or an experimentally validated cross-species boundary.

Keywords: DNA-damage response; observational resolution; qualitative biological networks; model comparison; Petri nets; weak bisimulation.

1. Introduction

Qualitative network models make biological hypotheses executable without requiring a fully parameterized kinetic description. Models assembled for related functions may nevertheless differ in organism, curation, initial condition, or update convention. For computational biology, comparing them requires more than recognizing similar wiring: one must ask whether the models can reproduce the same biologically relevant steps and choices under a stated readout.^{1–3}

DNA-damage response provides a concrete example. Animal/human and *Arabidopsis thaliana* models share lesion detection, checkpoint activation, repair, and recovery, but encode different regulatory and terminal responses. Mammalian p53-

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associated control and apoptosis are not the plant-specific SOG1 network, SMR-associated arrest, or damage-induced differentiation/endoreduplication. These distinctions are supported independently of our comparison by DDR literature.^{4–9} Asking simply whether the models are “similar” conceals the choice of biological resolution. We instead ask: *at which declared observational resolution do their encoded damage-response choices cease to correspond?*

A structural comparator such as PN-GDDA characterizes local Petri-net wiring, not transition labels, the initial marking, hidden events, or recursive reachable choices.¹⁰ Linear event sequences add order but still do not specify when a choice becomes committed. A model that commits early to one of two outcomes can have the same sequences as a model that keeps both outcomes available longer, yet fail to reproduce the latter’s branching options. This matters when a model comparison is intended to preserve choices, not merely lists of possible endpoints.

Our computational approach first declares what the biological readout can distinguish. We then construct the finite executable states and ask whether each observed step can be matched while preserving that ability at the resulting state pair. Matching in both directions using one consistent relation is weak bisimulation; matching in one direction is weak simulation. Internal events are ignored only when the declared interface makes them unobservable. These standard relations^{11,12} supply a branching-sensitive model comparison, rather than a new biological similarity score. For models N_1, N_2 and observational interface Σ ,

$$(N_1, N_2, \Sigma) \mapsto (\mathcal{L}(N_1), \mathcal{L}(N_2)) \mapsto C_\Sigma(\mathcal{L}(N_1), \mathcal{L}(N_2)), \quad (1.1)$$

where $\mathcal{L}(N_i)$ is the reachable labeled transition system and C_Σ reports the strongest supported relation. The biological question and the observational resolution precede that calculation (Fig. 1).

The principal result concerns the curated GIM (genome-instability/DNA-repair) pair. Its unchanged structures score 0.9976 by PN-GDDA, yet the behavioral answer changes across three declared interfaces. Coarse functional and pathway-resolved observations retain weak bisimilarity. Exposing distinct terminal mechanisms makes the models non-comparable in both directions. The useful output is the location of this change within the tested interfaces, not the rediscovery of the mechanistic differences already encoded in them.

The contributions are a resolution-specific comparison question, its executable GIM demonstration, a branching-sensitive computational method with explicit orientation, and independent formal/software checks supporting that method. Secondary curated comparisons illustrate one-way matching; an exploratory HPN-DREAM/CASPOTS analysis confronts model relations with held-out response data.^{13,14} The study does not establish organism-level equivalence, experimental validation of the GIM boundary, or predictive discrimination. It distinguishes the biological knowledge used to construct interfaces from the model-dependent relationship computed under them.

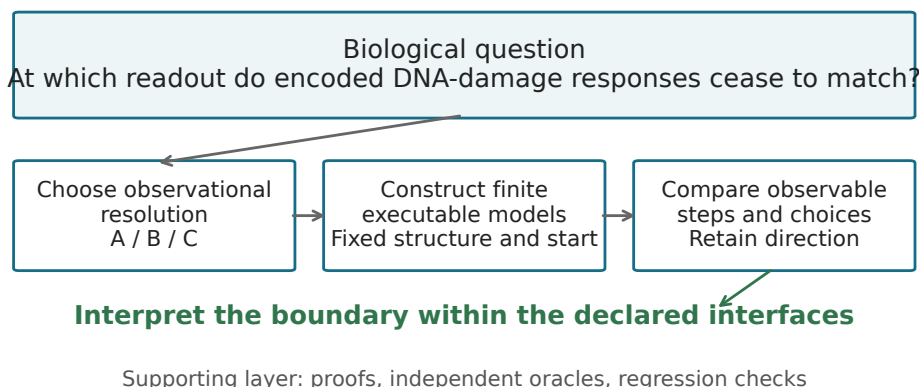


Fig. 1. Question-first comparison workflow. Biological readout selection determines the observational interface before executable models are compared. Formal verification supports the computation; the endpoint is an explicitly model-conditional interpretation of the compatibility boundary.

2. Biological comparison problem and models

2.1. Biological question and observational-interface principles

The central question is whether the animal/human and *Arabidopsis thaliana* DNA-damage models encode a common executable control organization, and at which observational resolution that correspondence ceases to hold. Independent biology supports comparing broad functions but also requires preserving mechanistic distinctions. ATM- and ATR-associated sensing, checkpoint control, repair, and recovery occur in both kingdoms; mammalian p53/apoptosis and the plant-specific SOG1, WEE1, SMR, and damage-associated endoreduplication programs are not interchangeable mechanisms.^{4–9,15,16}

We therefore evaluated three biologically motivated interfaces on exactly the same two Petri-net structures and initial markings. Interface A observes five coarse functions: damage detection, checkpoint activation, repair, restoration, and exit from the proliferative cycle. Interface B additionally distinguishes double-strand-break and replication-stress detection, ATM- and ATR-associated signaling, and broad repair families, while retaining a common terminal `cell_cycle_exit` readout. Interface C preserves those distinctions and separately exposes mammalian `apoptosis`, plant `smr_induction`, and the plant net’s combined `differentiation_or_endoreduplication` transition. The last label is a property of this compact encoding: the model does not infer differentiation and endoreduplication as separate branches. Table 1 records why each process is observed or hidden. The executable definitions and the literature keys were fixed in source data and are exported as a machine-readable audit table.

The interfaces are curated functional abstractions, not experimentally calibrated

Table 1. Biological rationale for the GIM observational interfaces. “A” denotes coarse function; “B/C” denotes the pathway-resolved interfaces.

Process	Animal/human encoding	<i>Arabidopsis thaliana</i> encoding	Interface treatment and independent support
Damage sensing	ATM-dominant DSB and ATR-dominant replication-stress channels	ATM/ATR channels upstream of SOG1	A merges detection; B/C distinguish the initiating channels. ^{5, 6, 9}
Checkpoint	CHK1/2–p53–p21 control	SOG1-regulated WEE1 and SMR control	A exposes function; B/C expose ATM/ATR but keep downstream relays internal. ^{4, 7, 8, 15}
Repair	DSB and excision-repair families followed by recovery	Broad DSB and excision-repair families followed by recovery	A merges repair; B/C separate repair families. ^{4, 6, 16}
Unresolved damage	Apoptotic exit from the proliferative pool	SMR-associated arrest followed by a combined differentiation/endoreduplication outcome	A/B collapse terminal loss; C distinguishes encoded mechanisms. ^{4, 8, 9, 17}

observation operators. Their order was fixed for this comparison from known mechanisms; it does not constitute a blinded prediction. The plant and animal nets are small safe interleaving encodings, not complete DDR pathways.

3. Computational comparison method

3.1. Labeled Petri nets and induced transition systems

Definition 1 (Labeled place/transition net). A labeled place/transition net is a tuple

$$N = (P, T, \text{pre}, \text{post}, \ell, m_0), \quad (3.1)$$

where P and T are finite sets of places and transitions, $\text{pre}, \text{post} : T \rightarrow \mathbb{N}^P$ are the input and output multisets, $m_0 \in \mathbb{N}^P$ is the initial marking, and $\ell : T \rightarrow \Sigma \cup \{\tau\}$ assigns either an observable action in the declared interface Σ or the internal action τ .

Under standard interleaving semantics,¹⁸ t is enabled at m when $\text{pre}(t) \leq m$ componentwise; firing it changes m to $m - \text{pre}(t) + \text{post}(t)$. The implementation uses a worklist to enumerate reachable markings explicitly. A state cap and a token bound make failure to establish a finite state space explicit rather than silently truncating the relation calculation. All curated Petri nets used in Section 2 are safe, but the definitions below apply to every finite reachable system.

Definition 2 (Finite labeled transition system). A finite labeled transition

system (LTS) is

$$L = (S, \Sigma_\tau, \longrightarrow, s_0), \quad \Sigma_\tau = \Sigma \cup \{\tau\}, \quad (3.2)$$

where S is a finite state set, $s_0 \in S$, and $\longrightarrow \subseteq S \times \Sigma_\tau \times S$. We write $s \xrightarrow{a} s'$ for $(s, a, s') \in \longrightarrow$. The reachable semantics of the net in Definition 1 is denoted by $\mathcal{L}(N)$.

3.2. Observable and weak transitions

Definition 3 (Silent closure and weak target sets). For $s \in S$, let

$$\text{Cl}_\tau(s) = \{u \in S : s \xrightarrow{\tau^*} u\}. \quad (3.3)$$

For $a \in \Sigma$, define

$$W_a(s) = \{v : \exists u, w \quad s \xrightarrow{\tau^*} u \xrightarrow{a} w \xrightarrow{\tau^*} v\}, \quad W_\tau(s) = \text{Cl}_\tau(s). \quad (3.4)$$

We write $s \xRightarrow{a} v$ when $v \in W_a(s)$.

The convention $W_\tau = \text{Cl}_\tau$ is important: a direct silent move on one side may be matched by zero or more silent moves on the other. It is also the convention implemented by the Python engine and used in the AUT exports checked by mCRL2.

3.3. Simulation and bisimulation

Let $L_i = (S_i, \Sigma_\tau, \longrightarrow_i, s_i^0)$ for $i \in \{1, 2\}$.

Definition 4 (Weak simulation). A relation $R \subseteq S_1 \times S_2$ is a weak simulation from L_1 to L_2 if, whenever $(x, y) \in R$ and $x \xrightarrow{a}_1 x'$, there is $y' \in W_a^2(y)$ such that $(x', y') \in R$. We write

$$L_1 \preceq_w L_2 \quad (3.5)$$

if (s_1^0, s_2^0) belongs to some such relation. Thus the right-hand system L_2 simulates the left-hand system L_1 ; the left-hand behavior is contained in the right-hand behavior under the interface.

Definition 5 (Weak and strong bisimulation). A relation $R \subseteq S_1 \times S_2$ is a weak bisimulation if it satisfies the condition in Definition 4 and, for every $(x, y) \in R$ and $y \xrightarrow{a}_2 y'$, there is $x' \in W_a^1(x)$ with $(x', y') \in R$. The initial systems are weakly bisimilar, written $L_1 \sim_w L_2$, when their initial pair belongs to such a relation. Strong bisimulation, written $L_1 \sim_s L_2$, replaces every weak target set $W_a^i(x)$ by the direct target set $\{x' : x \xrightarrow{a}_i x'\}$, including for $a = \tau$.

These are standard branching-time relations.^{11,12} The explicit orientation in (3.5) prevents the common ambiguity between “simulates” and “is simulated by” in one-way results.

3.4. Weak trace language

Definition 6 (Exact and truncated weak traces). The finite weak observable language of L is

$$L_w(L) = \{a_1 \cdots a_r \in \Sigma^* : s_0 \xRightarrow{a_1} \cdots \xRightarrow{a_r} s \text{ for some } s \in S\}. \quad (3.6)$$

Its depth- k restriction is $L_{\leq k}(L) = \{u \in L_w(L) : |u| \leq k\}$. The diagnostic distance is

$$d_k(L_1, L_2) = 1 - \frac{|L_{\leq k}(L_1) \cap L_{\leq k}(L_2)|}{|L_{\leq k}(L_1) \cup L_{\leq k}(L_2)|}. \quad (3.7)$$

The implementation computes the prefix-closed truncated sets in Definition 6. The statement $d_k = 0$ is therefore not identified with exact weak-trace equivalence. Exact weak-trace results come from mCRL2, acyclic path enumeration, or finite subset-product exploration without a depth cutoff; none is inferred from a truncated distance.

3.5. Graded behavioral classification

Definition 7 (Graded classifier). For finite L_1, L_2 over the declared interface Σ , define $C_\Sigma(L_1, L_2)$ by the following priority:

$$C_\Sigma(L_1, L_2) = \begin{cases} \text{strong,} & L_1 \sim_s L_2, \\ \text{weak,} & L_1 \not\sim_s L_2 \text{ and } L_1 \sim_w L_2, \\ \text{mutual,} & L_1 \not\sim_w L_2, L_1 \preceq_w L_2, L_2 \preceq_w L_1, \\ \text{left_in_right,} & L_1 \preceq_w L_2, L_2 \not\preceq_w L_1, \\ \text{right_in_left,} & L_2 \preceq_w L_1, L_1 \not\preceq_w L_2, \\ \text{noncomparable,} & \text{otherwise.} \end{cases} \quad (3.8)$$

The output is a reporting class, not a newly proposed equivalence relation.

Proposition 1 (Exclusive reporting). *The classifier in Definition 7 assigns exactly one class to every ordered pair of finite LTSs.*

Proof. The first two cases are selected before any simulation-only case. If neither is selected, the two simulation predicates have exactly four Boolean combinations: both true, only the first true, only the second true, or both false. These are the last four cases of (3.8), respectively. The cases are exhaustive and disjoint by their stated guards. $\square$

3.6. Computational implementation

Algorithm 1 summarizes the executable path. Reachability and the formal relations are exact whenever the finite state space is completed below the declared state and

token caps. The distance d_k , PN-GDDA, LTS-GDA, and the structural profile are diagnostics rather than formal relations.

Algorithm 1 (Executable comparison pipeline).

- (i) Enumerate the reachable LTS from the initial marking or logical state by a worklist; stop with an explicit error if a cap is exceeded.
- (ii) Cache $\text{Cl}_\tau(s)$ and construct $W_a(s)$ by collecting a -successors from the closure and the closures of their targets.
- (iii) Initialize $R = S_1 \times S_2$ and delete a pair immediately when it fails the selected transfer operator; repeat complete passes until none is deleted.
- (iv) Evaluate strong and weak bisimilarity and both simulation directions, apply Eq. (3.8), and attach trace and structural diagnostics.

The finite relation-deletion theorem and the strict hierarchy underlying this reporting priority are proved in Appendix A. Example 1 provides an explicit simulation witness and a failed-successor proof for equal traces with one-way simulation. PN-GDDA, LTS-GDA and trace distances answer complementary diagnostic questions; their definitions and implementation checks are retained in Supplementary Validation, Section S1.

4. Main result: a resolution boundary in GIM

4.1. Structure held constant

The animal and plant nets reach 13 and 14 states, respectively. No place, transition, arc, or initial marking changes across the three interfaces. Native PN-GDDA is 0.9975846 throughout (0.9975175 with the independently derived 576-orbit partition). This describes nearly identical selected local wiring, not mechanistic identity or execution equivalence. Figure 2 makes the comparison’s biological assumptions and computed outcomes visible together.

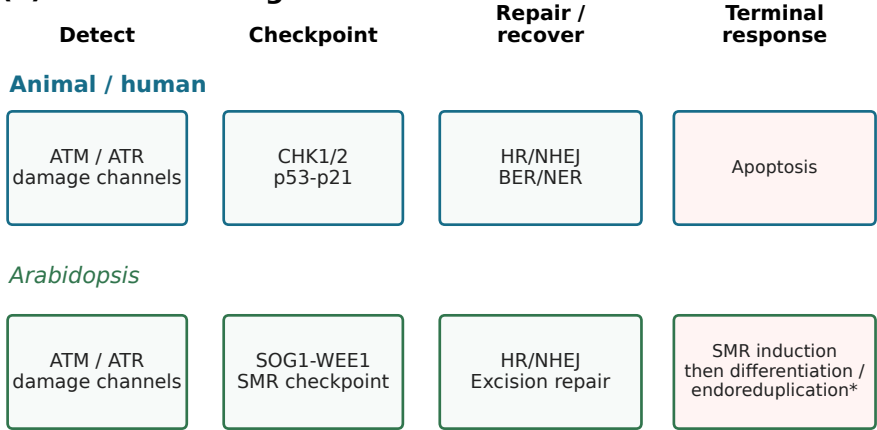
4.2. Coarse function retains correspondence

At Interface A, every observable detection, checkpoint, repair, restoration, or cycle-exit step in either executable model can be matched by the other, with this matching preserved at the resulting states. Internal relays can require different numbers of steps. Formally, the systems are weakly but not strongly bisimilar. The result licenses comparison at the declared shared functional level; it does not identify their molecular implementations.

4.3. Pathway distinctions do not yet break the relation

Interface B distinguishes damage channels, ATM/ATR-associated signaling, and broad repair families. The models still match recursively when terminal loss of proliferative capacity remains a common observation. Both simulation directions and

(a) Encoded biological functions and mechanisms



*One combined terminal transition in the plant encoding, not two alternatives.

(b) Same structures, different observational resolution

Readout	What is distinguished	PN-GDDA	Branching relation
A Coarse	Shared functions; terminal loss of proliferative capacity	0.9976	Weakly bisimilar
B Pathway	Damage, ATM/ATR and repair; terminal mechanisms collapsed	0.9976	Weakly bisimilar
C Terminal	Pathway events plus separate animal / plant terminal mechanisms	0.9976	Non-comparable (both directions fail)

Model- and interface-conditional results; no organism-level equivalence

Fig. 2. Central GIM comparison. Upper-panel columns align encoded animal/human and *Arabidopsis thaliana* functional categories; they are not a causal pathway or a required sequence of outcomes. The lower panel shows what each declared interface observes and the computed branching relation. PN-GDDA remains 0.9976 because the structures are unchanged. The combined plant terminal transition does not distinguish differentiation from endoreduplication. Shared labels at A/B express a functional abstraction, not identity between p53 and SOG1 or between terminal mechanisms.

weak bisimilarity hold; strong bisimilarity does not. Thus, among these encodings and interfaces, distinguishing those pathway events alone is insufficient to eliminate the correspondence.

Table 2. Recomputed GIM results. $\text{Animal} \preceq \text{Plant}$ means that the plant model simulates the animal model; $\text{Plant} \preceq \text{Animal}$ is the reverse. Strong bisimilarity is false throughout.

Interface	$\text{Animal} \preceq \text{Plant}$	$\text{Plant} \preceq \text{Animal}$	Class	d_6	PN-GDDA
A: coarse	Y	Y	weak bisimulation	0.0000	0.9976
B: pathway	Y	Y	weak bisimulation	0.0000	0.9976
C: terminal	N	N	non-comparable	0.3529	0.9976

4.4. Terminal mechanisms eliminate both directions

At Interface C, mammalian `apoptosis` and plant `smr_induction` followed by `differentiation_or_endoreduplication` become distinct observations. Neither model can match all observable moves of the other, so weak bisimilarity and both simulation directions fail. This is not an unexpected biological discovery: the terminal labels intentionally encode prior knowledge of different mechanisms. The computational result confirms where, in the tested sequence of readouts, that distinction prevents recursive matching. The depth-six trace distance becomes 0.3529 rather than zero; LTS-GDA changes from 0.9508 to 0.8921. These diagnostics accompany the formal decision, not replace it (Table 2).

4.5. What changes when observability changes?

Interfaces A and B aggregate terminal mechanisms into a shared functional readout, whereas C separates them. The observed boundary is therefore *between B and C within this declared interface family*, not a uniquely identified or globally minimal biological resolution. There is no exhaustive search over all possible interfaces, and no experimental threshold is inferred. The constant structural score isolates observability from topological change.

Prior biology supplies the distinct terminal mechanisms; PN-GDDA describes their selected wiring; the executable comparison determines whether a recursive correspondence survives each declared readout. Whether such a model-level boundary predicts a cross-species experimental boundary remains unvalidated. One testable interpretation is that readouts aggregating terminal loss of proliferative capacity may preserve an apparent correspondence that mechanism-specific readouts do not. This is a model-derived hypothesis, not an assay result. The full Petri nets remain in Appendix B.

5. Supporting curated comparisons

Five compact plant–animal pairs encode DNA repair/genome instability (GIM), energy metabolism (DCE), cell-cycle control (SPS), cell death/autophagy (RCD), and immune control (AID). Comparative studies and cancer-hallmark literature guided component selection; dedicated reviews informed plant DNA-damage, cell-cycle, and autophagy mechanisms.^{6,19–21} Transition-level provenance is machine-checked,

Table 3. Supporting curated model-pair outputs. GIM is reported at Interface A; all relations and d_6 values are interface-conditional, whereas PN-GDDA is structural.

Module	Process	Formal class	d_6	PN-592	Orbit-576
GIM	DNA repair	weak bisimulation	0.000	0.9976	0.9975
DCE	energy metabolism	weak bisimulation	0.000	0.9969	0.9968
SPS	cell-cycle control	weak bisimulation	0.000	1.0000	1.0000
RCD	death/autophagy	plant in animal	0.143	1.0000	1.0000
AID	immune control	non-comparable	0.429	0.9907	0.9904

which establishes where encoded assumptions came from and whether documentation is complete. It does not establish that the nets reproduce independent experiments. These manually curated models contain no kinetics, concentrations, tissue context, or independent experimental calibration.

The ten nets produced 5–14 reachable states. The GIM row in Table 3 reports Interface A; GIM, DCE, and SPS were weakly but not strongly bisimilar; RCD gave plant-in-animal one-way simulation; AID was non-comparable. Table 3 juxtaposes these classes with native PN-GDDA. The 592- and 576-orbit variants changed no score by 0.001 and no threshold decision: all five pairs were structurally similar at 0.9 despite three different behavioral classes.

RCD supplies a secondary directional result: both encoded nets contain stress, mTOR inhibition, autophagy, and survival, but only the animal encoding exposes a canonical caspase-apoptosis branch; the plant branch uses an internal metacaspase event. The plant LTS is weakly simulated by the animal LTS, whereas the reverse direction fails. This means containment only for the declared labels of these two models; it does not equate plant programmed cell death with mammalian apoptosis. Silent subdivision preserved every class over 300 refinements per module, while targeted relabeling changed the expected relations. Distances against 2,000 label permutations assess internal interface sensitivity, not external biological significance. DCE, SPS, and AID are retained as supporting model-ingestion and relation-diversity examples rather than independent biological validation.

6. Exploratory confrontation with HPN-DREAM data

An independent-data check is retained as a secondary, exploratory analysis, not as validation of the GIM resolution boundary. Public HPN-DREAM/CASPOTS families contain 72 BT20, 191 BT549, and 21 MCF7 Boolean networks. Structure-only medoid selection, fixed before test-response access, chooses zero-based rows 24, 137, and 9.^{13,14} The common interface contains seven measured PI3K/MAPK/mTOR readouts under three shared held-out mTOR-inhibitor conditions. Initial values, clamping, and update conventions are specified in Supplementary Validation, Section S3.

The nine asynchronous pair-condition comparisons contain six one-way simulations and three non-comparable pairs. The six directions comprise three left-model-

in-right-model and three right-model-in-left-model relations, not positions on an ordinal similarity scale. Direction is preserved in the result fields; no scalar formal-class test is performed. Median profile RMSE is 0.2757, 0.1919, and 0.2071 in those three nominal categories, respectively, with only three observations per category. Synchronous updating preserves seven of the nine classes. These summaries cannot establish predictive value.

The retained continuous diagnostics are inconclusive: truncated-trace distance has Spearman $\rho = -0.596$, exact one-sided $p = 0.556$; LTS-GDA distance has $\rho = 0.669$, $p = 0.097$, using all 216 condition-stratified permutations. The CASPOTS native-context advantage over cross-cell medoids is 0.00212 excess-RMSE units and is not established by the exact three-cell sign test ($p = 0.25$). No more favorable statistic was substituted. The full methods, negative results, and plot remain in Supplementary Validation, Section S3; they show a route to data confrontation without claiming prognosis, native cell-line specificity, or experimental equivalence.

7. Supporting formal and implementation checks

Definitions and the executed relation-deletion method are supported by the proofs in Appendix A. Table 4 states the checks, independent bases, and observed outcomes in the main article. These checks address whether the implemented relations and structural score behave as specified; they do not establish biological validity.

Table 4. Main implementation-verification results. Agreement refers to the declared inputs and predicates, not predictive accuracy on biological data.

Check	Independent basis and scope	Observed result
Constructed cases	Six pairs with classes fixed before classification	6/6 classes and 6/6 binary weak-equivalence decisions recovered.
External equivalence	mCRL2 on six synthetic, six public-model controls, and nine HPN pairs	42/42 strong/weak-bisimilarity decisions agree.
Directional audit	Raw-edge attacker-defender game on 32 model pairs and three hierarchy witnesses	No disagreement on either weak-simulation direction or strong/weak bisimilarity.
Exhaustive finite check	All 260 one-/two-state LTSs over $\{a, \tau\}$; 67,600 ordered pairs	No simulation or weak-bisimulation discrepancy; no hierarchy or trace-inclusion violation.
Structural score	Holmes 1.1.1: 151 graphlets, 592 slots; four-node net and one-arc-deletion control	PN-GDDA agrees to 12 decimal places; absolute difference 3.33×10^{-16} .

The public cell-cycle and p53-Mdm2 controls test SBML-qual and GINML

imports; all six copy, silent-refinement, and label-perturbation controls recover their predeclared outcomes.^{22–25} Holmes supplies an external reference for the PN-GDDA implementation.²⁶ mCRL2 supplies the external equivalence checks.²⁷ Its simulation-preorder checks are used only for the two tau-free hierarchy witnesses, where strong and weak simulation coincide; no general weak-preorder verification is attributed to that tool.

For directional checks, the independent game constructs weak targets directly from raw edges without calling the production closure or relation routines. The 32 pairs comprise six synthetic, three GIM-interface, five curated, and 18 asynchronous/synchronous HPN comparisons. The exhaustive universe fixes initial state zero and enumerates every edge subset; it is not a proof for arbitrary finite systems. In particular, Example 1 separately establishes early-to-late simulation, failure of the reverse, and exact trace equality using an explicit witness and a failed-successor argument. No exact trace conclusion is inferred from a truncated distance.

The evidence layers remain distinct: proofs establish mathematical claims; tests establish implementation agreement on tested inputs; model provenance and import checks establish executable assumptions; biological literature and held-out data support or limit biological interpretation. Formal agreement alone does not calibrate a curated net against experiments. Supplementary Validation S1–S2 and S4 retain per-case outputs, comparator definitions, tool procedures, and scaling details; the principal GIM evidence remains in Section 4.

8. Discussion

8.1. *What does structural similarity tell us?*

PN-GDDA describes similarity of local directed bipartite Petri-net wiring. The GIM value 0.9976 is therefore meaningful but deliberately incomplete as a statement about executable behavior. It remains constant when labels change because the underlying structure does not. A label-blind score cannot determine labeled trace equality or simulation in general (Proposition 4); this is a boundary of the question it answers, not a defect in the comparator. Structural and behavioral comparisons are complementary, rather than competitors for one universal notion of similarity.

8.2. *What does behavioral comparison add?*

The executable comparison determines whether observable steps can be matched recursively from the declared initial states. Unlike a trace list, it retains the location at which choices become committed (Example 1). This distinction is relevant when transferring qualitative model behavior requires preserving alternative continuations rather than merely permitting the same sequences. Related formal work on asynchronous Boolean networks also emphasizes reachable branching in phenotypic analysis.²⁸

In GIM, the model-level answer is correspondence at A/B and its loss at C. In RCD, only the plant model’s declared branching behavior is reproducible by

the animal model, not conversely. Simulation is stronger than saying that each plant trace occurs in the animal model: every related successor pair must continue to satisfy the matching condition. Neither result transfers to other interfaces or organisms without re-analysis.

8.3. *What does interface refinement mean biologically?*

An observational interface states which differences a biological readout is allowed to retain, merge, or hide. At A/B, the GIM terminal responses are collapsed into loss of proliferative capacity; at C, their encoded mechanisms are distinguishable. Their incompatibility at C is consequently consistent with the model construction, not independent evidence that the method has discovered a new organism-level difference. The contribution is to determine the survival or failure of recursive correspondence for the full executable models under each explicit interface.

The result localizes a boundary within three declared readouts; it does not optimize over interfaces or identify a unique minimal observational scale. Because the interfaces were informed by known mechanisms, the C result is not a blinded prediction. A prospective test would need independently selected readouts, matched biological conditions, and outcomes not used in model or interface construction. The present assay interpretation remains a hypothesis. The distinction between a combined differentiation/endoreduplication transition and two independently selectable outcomes is also essential: only the combined transition is encoded here.

8.4. *Execution assumptions and evidence limits*

The comparison depends on transitions, initial states, labels, and update semantics, not on graph structure alone. Asynchronous versus synchronous execution changes public-model reachability and two HPN classes. General asynchronous, most-permissive, and stochastic semantics were not evaluated.²⁹ Weak bisimulation here is divergence-insensitive and concerns untimed interleaving behavior, not rates, probabilities, or true concurrency. Controlled silent refinement preserves the relation only under the stated construction (Proposition 3), not under arbitrary insertions of conflicts or observable branches.

The GIM pair is manually curated and compact, with no kinetics, concentrations, tissue context, or independent experimental calibration. Literature justifies the chosen biological distinctions but does not certify the entire executable model. HPN provides independent measurements yet has only nine pair-condition observations, three cell lines, and inconclusive associations. Its formal directions are nominal; assigning them an ordinal score would lose information. CASPOTS RMSE is minimum-distance compatibility over admissible trajectories, not a unique time-series prediction. These limitations cannot be removed by adding more software controls.

8.5. Computational scope

Explicit reachability remains vulnerable to state-space explosion, and a candidate relation stores the product of the two state sets. The scaling study in Supplementary Validation is illustrative, not evidence of genome-scale performance. Applying native PN-GDDA to logical rules would additionally require a non-unique Petri-net translation. Larger independently curated model pairs, prospectively chosen interfaces, and empirical readouts are needed before claiming wider biological utility. The present method is most defensible as a reproducible, question-specific comparison of explicit qualitative models.

9. Conclusion

Two curated DNA-damage response models with constant PN-GDDA similarity 0.9976 retain recursive behavioral correspondence at coarse and pathway resolution, but become non-comparable when their encoded terminal mechanisms are separately observed. For these models, correspondence depends on observational resolution; the boundary lies between Interfaces B and C within the declared family.

The computational contribution is to make that conditional boundary explicit and reproducible, while preserving the difference between similar wiring, matching traces, and matching branching. Proofs and independent checks support the formal decisions without substituting for biological evidence. The result does not establish organism-level equivalence, mechanistic identity, or prognosis. Its empirical relevance remains to be tested beyond these curated models; the secondary HPN analysis is exploratory and inconclusive.

Appendix A. Mathematical properties and quantifier audit

For $a \in \Sigma_\tau$, let $D_a^i(x) = \{x' : x \xrightarrow{a}_i x'\}$ and $U = S_1 \times S_2$. Define operators on $\mathcal{P}(U)$ by

$$\Phi_{\leq}(R) = \{(x, y) \in U : \forall a, x' \in D_a^1(x) \exists y' \in W_a^2(y), (x', y') \in R\}, \quad (\text{A.1})$$

$$\Phi_w(R) = \{(x, y) \in \Phi_{\leq}(R) : \forall a, y' \in D_a^2(y) \exists x' \in W_a^1(x), (x', y') \in R\}, \quad (\text{A.2})$$

$$\begin{aligned} \Phi_s(R) = \{(x, y) \in U : \forall a, x' \in D_a^1(x) \exists y' \in D_a^2(y), (x', y') \in R, \\ \forall a, y' \in D_a^2(y) \exists x' \in D_a^1(x), (x', y') \in R\}. \end{aligned} \quad (\text{A.3})$$

Theorem 1 (Correctness of relation deletion). *For each $\Phi \in \{\Phi_{\leq}, \Phi_w, \Phi_s\}$, the operator is monotone. The implemented algorithm starts with $R = U$, repeatedly deletes any currently examined pair $z \notin \Phi(R)$, and stops after a complete pass with no deletion. On finite LTSs it terminates and returns the greatest fixed point $\nu\Phi$. Consequently, testing membership of (s_1^0, s_2^0) computes, respectively, weak simulation in the orientation of Definition 4, weak bisimilarity, and strong bisimilarity.*

Proof. If $R \subseteq Q$, every witness pair in R remains available in Q ; hence $\Phi(R) \subseteq$

Table 5. Distinct witnesses for the three failed converses. All pairs have equal exact weak-trace languages; $X \preceq Y$ means Y simulates X .

Converse refuted	Witness pair (X, Y)	Decisive properties
Weak implies strong bisimilarity	$a.0, a.\tau.0$	weak, not strong
Mutual simulation implies weak bisimilarity	$a.(b+c) + a.b, a.(b+c)$	both directions, not weak
Exact trace equality implies mutual simulation	late choice, early choice	early $\preceq$ late only

$\Phi(Q)$ for each operator. Every deletion strictly decreases the finite relation, so termination requires at most $|U|$ deletions.

Let $G = \nu\Phi$. The invariant $G \subseteq R$ holds initially. If it holds before a deletion and $z \in G$, then $z \in \Phi(G) \subseteq \Phi(R)$, so z cannot be deleted. Thus G remains a subset of the terminal relation R_* . A final pass without deletion gives $R_* \subseteq \Phi(R_*)$. Conversely, each $z \notin R_*$ was deleted from some R_z with $z \notin \Phi(R_z)$. Since $R_* \subseteq R_z$, monotonicity implies $\Phi(R_*) \subseteq \Phi(R_z)$ and therefore $z \notin \Phi(R_*)$. Hence $R_* = \Phi(R_*)$. Because R_* is a fixed point, $R_* \subseteq G$; together with the invariant this yields $R_* = G$. The three operators are exactly the transfer clauses of Definitions 4 and 5. $\square$

Remark 1 (Resource bound). Writing $n_i = |S_i|$, the candidate relation stores $O(n_1 n_2)$ pairs and permits at most $n_1 n_2$ deletions and $n_1 n_2 + 1$ passes. Each pass scans current pairs, outgoing moves, and matching direct or weak targets. Silent closures are cached; visible weak targets are constructed on demand. We therefore report these parameters instead of claiming an unjustified tight time bound.

Proposition 2 (Hierarchy of behavioral relations). *For finite LTSs under Definitions 3–6,*

$$L_1 \sim_s L_2 \implies L_1 \sim_w L_2 \implies (L_1 \preceq_w L_2 \wedge L_2 \preceq_w L_1) \implies L_w(L_1) = L_w(L_2). \quad (\text{A.4})$$

The converses do not hold in general.

Proof. A direct match is a permitted weak match, and each direction of a weak bisimulation is a weak simulation. If $L_1 \preceq_w L_2$, induction along any finite concrete path matches observable moves by $\tau^* a \tau^*$ and silent moves by τ^* , preserving related successors. Erasing τ gives $L_w(L_1) \subseteq L_w(L_2)$; the reverse simulation gives equality. Proposition 3 and Example 1 refute the first and last converses. For the middle converse, compare $a.(b+c) + a.b$ with $a.(b+c)$: each simulates the other, but the $a.b$ successor cannot be bisimilar to a state offering both b and c . $\square$

For the middle row, write x_{bc}, x_b for the two a -successors of X and y_{bc} for the sole a -successor of Y . A simulation from X to Y relates both x_{bc} and x_b to y_{bc} , with corresponding terminal pairs. A simulation from Y to X relates y_{bc} only to x_{bc} . Both initial pairs therefore survive. A bisimulation would additionally need

to match X 's transition to x_b with y_{bc} ; the latter's c -move has no counterpart at x_b . This witness concerns only the middle converse, not the trace-equality converse refuted by Example 1.

Proposition 3 (Controlled silent edge refinement). *Let L contain $p \xrightarrow{a} q$ with $a \in \Sigma$ and $p \neq q$. Construct L' by copying all original states and transitions except this edge, and replacing it by*

$$\bar{p} \xrightarrow{a} r \xrightarrow{\tau} \bar{q}, \quad (\text{A.5})$$

where r is fresh and has no other incident behavior. Then $L \sim_w L'$. Repeated refinements of this form also preserve weak bisimilarity. If p and $\bar{p}$ are initial, the selected edge is the only a -edge from $\bar{p}$ in L' , and every a -successor of p in L has no outgoing τ transition, then $L \not\sim_s L'$.

Proof. Let $R = \{(x, \bar{x}) : x \in S\} \cup \{(q, r)\}$. For every copied pair other than the source, direct moves are copied. At $(p, \bar{p})$, the original edge is matched by $\bar{p} \xrightarrow{a} r \xrightarrow{\tau} \bar{q}$, while $\bar{p} \xrightarrow{a} r$ is matched by $p \xrightarrow{a} q$ and ends in (q, r) . At (q, r) , moves from q are matched after $r \xrightarrow{\tau} \bar{q}$; that silent move is itself matched by zero silent moves at q . Hence R is a weak bisimulation, and repetition follows by transitivity. Under the additional conditions, a strong match for $\bar{p} \xrightarrow{a} r$ must relate r to an a -successor x of p , but $r \xrightarrow{\tau} \bar{q}$ would require the excluded direct τ -move from x . Thus strong bisimilarity fails. $\square$

Proposition 4 (Limit of label-blind structural comparison). *Let M be any comparator determined only by unlabeled structure and invariant under relabeling of transitions. Then M cannot characterize weak-trace equality, weak simulation or weak bisimilarity for labeled systems in general.*

Proof. Take two systems with the same two states and one directed edge. Label the edge a in L_a and $b \neq a$ in L_b . Their unlabeled structures are identical, so $M(L_a, L_b) = M(L_a, L_a)$. Yet their languages are $\{\epsilon, a\}$ and $\{\epsilon, b\}$, and neither observable move can be matched by the other system. Thus no decision based only on M can characterize the stated labeled relations. $\square$

Proposition 4 does not identify a defect in PN-GDDA. It states that a label-blind structural method answers a different question and cannot substitute for a labeled behavioral relation.

Example 1 (Equal traces, different branching). Let the late-choice system start at ℓ_0 and have exactly

$$\ell_0 \xrightarrow{a} \ell_1, \quad \ell_1 \xrightarrow{b} \ell_b, \quad \ell_1 \xrightarrow{c} \ell_c, \quad (\text{A.6})$$

and let the early-choice system start at e_0 and have exactly

$$e_0 \xrightarrow{a} e_b, \quad e_0 \xrightarrow{a} e_c, \quad e_b \xrightarrow{b} e'_b, \quad e_c \xrightarrow{c} e'_c. \quad (\text{A.7})$$

All unlisted states are terminal; neither system has silent transitions or cycles. Enumerating all paths, whose lengths are at most two, gives both exact prefix-closed languages as $\{\epsilon, a, ab, ac\}$, not merely a truncated match.

An explicit simulation from early to late is

$$R_{E \rightarrow L} = \{(e_0, \ell_0), (e_b, \ell_1), (e_c, \ell_1), (e'_b, \ell_b), (e'_c, \ell_c)\}. \quad (\text{A.8})$$

Both initial a -moves are matched by $\ell_0 \xrightarrow{a} \ell_1$. At (e_b, ℓ_1) the sole b -move has a match; at (e_c, ℓ_1) the sole c -move has a match. Terminal pairs impose no further obligations. Thus $L_{\text{early}} \preceq_w L_{\text{late}}$.

Conversely, suppose a simulation R from late to early contains (ℓ_0, e_0) . Matching the single transition $\ell_0 \xrightarrow{a} \ell_1$ requires choosing one successor of e_0 and including the resulting pair in R . If $(\ell_1, e_b) \in R$, the c -move from ℓ_1 has no match. If $(\ell_1, e_c) \in R$, its b -move has no match. These exhaust the possible a -successors, so no such R exists: $L_{\text{late}} \not\preceq_w L_{\text{early}}$. Weak bisimilarity therefore also fails.

Different outgoing transitions from a related pair may indeed choose different matching successors. However, the *one* initial late a -transition must be matched by an early successor pair that itself satisfies every subsequent transfer obligation. Trace matching may select the e_b branch for ab and the e_c branch for ac independently; simulation cannot combine those two states into one defender state after the initial move. This is the quantifier distinction behind the final strictness claim of Proposition 2.

Appendix B. Detailed GIM Petri-net encodings

Figures 3 and 4 show the fixed executable nets; the three interfaces change only transition labels.

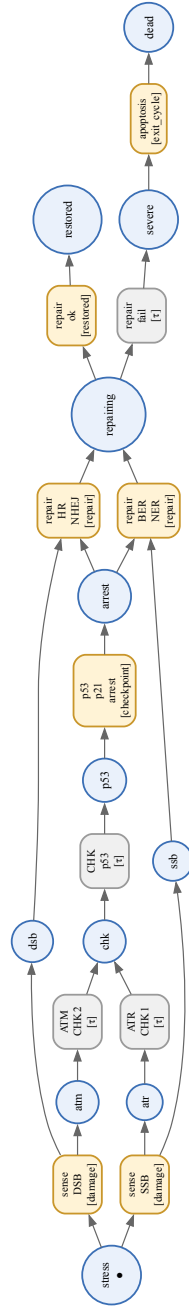


Fig. 3. Detailed animal/human GIM Petri net. Circles are places, rounded boxes are transitions, brackets show the base/coarse observable labels, and gray τ transitions are internal.

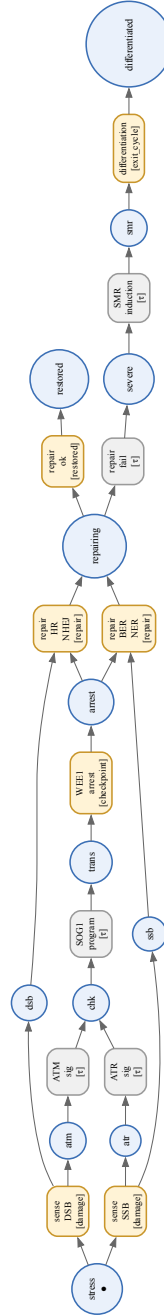


Fig. 4. Detailed *Arabidopsis thaliana* GIM Petri net, using the same graphical conventions as Fig. 3. The terminal transition is a combined outcome in this encoding and does not distinguish differentiation from endoreduplication.

Data and code availability

Code, curated definitions, hash-verified public models, normalized response files, metadata, and generated results are available under the MIT License at <https://github.com/cero1979/BisimulationMol>. The second-revision source and audits are in `revision_R2/`. The repository documents formal/software verification, interface definitions, figure regeneration, tests, and notebook execution. Machine-readable GIM and HPN results are provided as `results/gim_interface_sensitivity.csv` and `results/hpn_dream_formal_data_validation.csv`. No confidential or participant-level data were used.

Declarations

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